

Philosophy of Science as a Tool for Scientific Practice

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Content

1. Introduction to Philosophy of Science

2. “AI Ethics” as both an Epistemic and Ethical Concept: A Useful Lesson for Machine Learning?

presented by Marilou Niedda



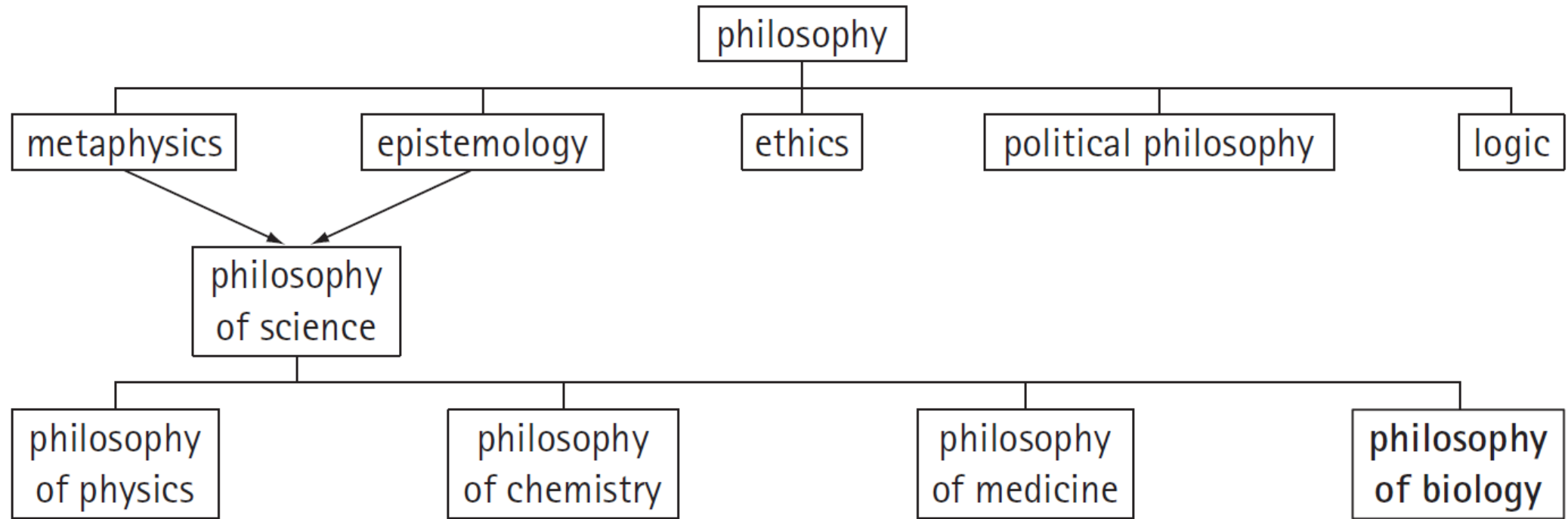


Figure 0.2: Philosophy of biology classified

Philosophy

Metaphysics: what kind of things there are (independent from human experience)

Epistemology: Concerns knowledge, how it is obtained, what is the difference between true knowledge and opinion.

Ethics: Normativity that prescribes human behavior usually in terms of rights, obligations, benefits to society, fairness, or specific virtue.

Political philosophy: Philosophical reflection on how best to arrange our collective life

Logic: Aims to distinguish good from bad reasoning by using symbols and logical rules. Premises must lead to logical conclusions.

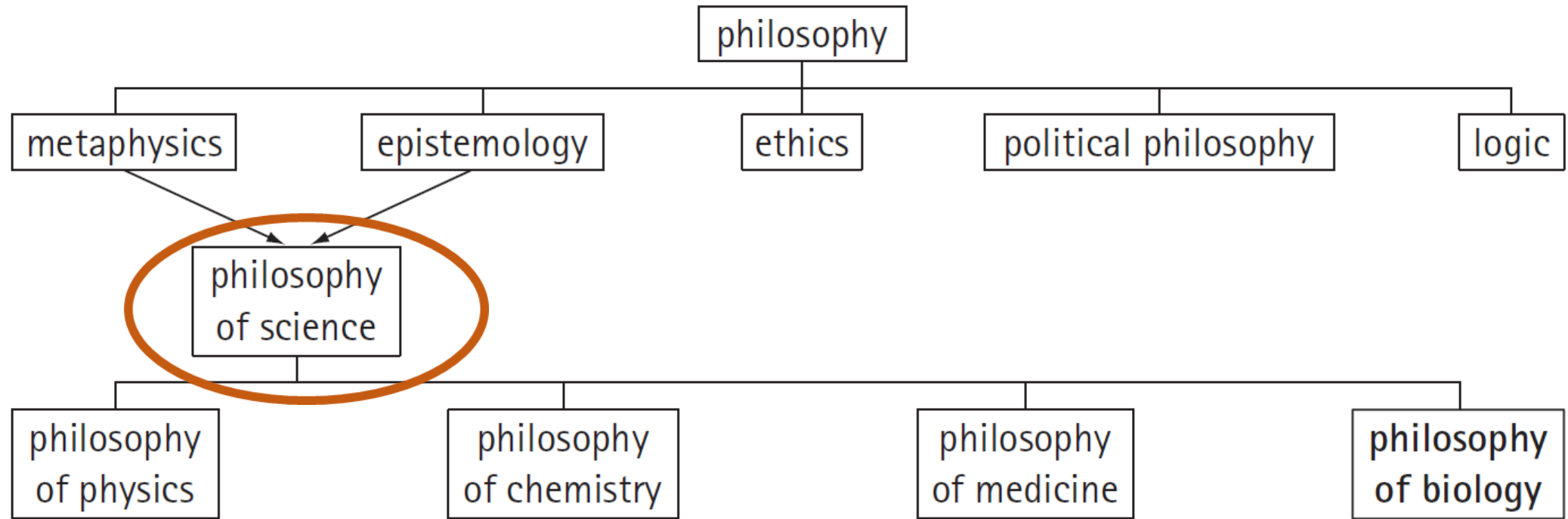


Figure 0.2: Philosophy of biology classified

Philosophy of science

- Traditional philosophy of science
- It studies science and its components, such as theories, models, laws of nature, empirical evidence and the relation of scientific knowledge to "truth".
- Epistemology, metaphysics, logics and ethics.
- Causality, realism, antirealism, truth, laws of nature...
- How can we distinguish science from non-science or pseudoscience? What is scientific knowledge? What constitutes good evidence?



Philosophy of science

From universal truth to situated knowledge

Situated knowledges are about communities, not about isolated individuals. The only way to find a larger vision is to be somewhere in particular (...)

Its images are not the products of escape and transcendence of limits (the view from above) but the joining of partial views and halting voices into a collective subject position. (Haraway 1988, 590)



Philosophy of science in practice

- Pays attention to how science is done in day to day (not idealized scenarios)
- Relies on empirical data gathered through observation, surveys, interviews...
- Acknowledges social actors and communities as important in defining scientific agreements in different disciplines: social epistemology
- Knowledge produced by groups of experts
- Groups of experts share theories, methods, practices and values



Philosophy of science in practice



- Investigates epistemic and non-epistemic values intrinsic to science
- **Epistemic values:** simplicity vs complexity, reproducibility, coherence, clarity, completeness, empirical adequacy, accuracy, consistency, scope, fruitfulness, explanatory power, objectivity, robustness...
- **Non-epistemic values:** promoting equity, social justice, safety, preventing harm, marketability, translatability to applications, aesthetic values, convenience, cost reduction...

What can history and philosophy do for science?

- To identify implicit assumptions in theories
- For conceptual clarification (what is fitness, what is a species, what is a cause, what is correlation)
- To identify epistemic and non-epistemic values or commitments
- To understand the effects of such values in knowledge
- To generate new research questions



Many thanks!

Research project:

Microbiome Research and Race in the 'Local South'



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“AI Ethics” as both an Epistemic and Ethical Concept: A Useful Lesson for Machine Learning?





Why should AI ethics matter to *you*?

How can you make your practice more
ethical?

1- 'AI ETHICS' as an ethics-cum-epistemology framework

Connecting ethics and epistemology of AI

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Abstract

The need for fair and just AI is often related to the possibility of understanding AI itself, in other words, of turning an opaque box into a glass box, as inspectable as possible. Transparency and explainability, however, pertain to the technical domain and to philosophy of science, thus leaving the ethics and epistemology of AI largely disconnected. To remedy this, we propose an integrated approach premised on the idea that a glass-box epistemology should explicitly consider how to incorporate values and other normative considerations, such as intersectoral vulnerabilities, at critical stages of the whole process from design and implementation to use and assessment. To connect ethics and epistemology of AI, we perform a double shift of focus. First, we move from trusting the output of an AI system to trusting the process that leads to the outcome. Second, we move from expert assessment to more inclusive assessment strategies, aiming to facilitate expert and non-expert assessment. Together, these two moves yield a framework usable for experts and non-experts when they inquire into relevant epistemological and ethical aspects of AI systems. We dub our framework 'epistemology-cum-ethics' to signal the equal importance of both aspects. We develop it from the vantage point of the designers: how to create the conditions to internalize values into the whole process of design, implementation, use, and assessment of an AI system, in which values (epistemic and non-epistemic) are explicitly considered at each stage and inspectable by every salient actor involved at any moment.

Keywords Glass-box AI · Ethics of AI · Epistemology of AI · Critical Questions · Holistic Model Validation · Ethics-cum-Epistemology

What matters is to *trust the process that leads to an AI's system outcome*, rather than blindly trusting the outcome

2- 'AI ETHICS' as a *feature importance* technique

Epistemic values in feature importance methods: Lessons from feminist epistemology

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ABSTRACT

As the public seeks greater accountability and transparency from machine learning algorithms, the research literature on methods to explain algorithms and their outputs has rapidly expanded. Feature importance methods form a popular class of explanation methods. In this paper, we apply the lens of feminist epistemology to recent feature importance research. We investigate what epistemic values are implicitly embedded in feature importance methods and how or whether they are in conflict with feminist epistemology. We offer some suggestions on how to conduct research on explanations that respects feminist epistemic values, taking into account the importance of social context, the epistemic privileges of subjugated knowers, and adopting more interactional ways of knowing.

KEYWORDS

explanation, philosophy, epistemology, machine learning, feature importance, feminism, methodology

inputs to a function. They are a popular way of approaching demands for explainability, as they fit the human tendency to attribute outcomes to specific factors.

As feature importance methods proliferate, practitioners struggle to decide which methods to use, and creators of methods strive to demonstrate that their methods are superior to others. However, much of the evaluation of various methods' pros and cons proceeds in an implicitly value-neutral context: rarely do participants in this discourse pause to consider if certain supposed theoretical or practical virtues would be virtues in all social or applicational contexts.¹ Papers introducing new feature importance methods state that certain properties of the proposed methods are "desirable" without indicating the contexts or uses for which those properties are desirable [51, 76]. Stating a property as "desirable" without further elaboration elides questions such as: Desirable to whom? For what purposes? When applied to what types of data?

In this paper, we argue that a number of popular feature impor-

"a feminist approach would use formalisms when appropriate, with a careful attention to what possibilities a particular formal framework excludes or renders unsayable, and what it inadvertently favours as prescriptive" (p.8)

Features choices are *always already* socially constituted & situated

In short

- 1/ We are all responsible for making AI systems better, but our levels of responsibility vary according to our level of expertise or our social role within societies
- 2/ For people with a technical background: use certain ML techniques aiming to do the least potential harm (a matter of 'social' probability) + some level of reflexivity